

Ali Raza

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PROFESSIONAL SUMMARY

Software Engineering student with hands-on experience in machine learning, backend development, and REST API development. Built end-to-end projects in regression, classification, and churn prediction using Python, pandas, and scikit-learn, alongside Django and REST API development. Seeking an ML Intern, Backend Developer, or AI Intern role to apply strong analytical and software engineering skills.

EDUCATION

Bachelor of Science in Software Engineering

CGPA: 3.68/4.00

The Islamia University of Bahawalpur

6th Semester

Relevant focus: machine learning, backend development, REST APIs, databases, and software engineering fundamentals.

SELECTED PROJECTS

ShortIQ — URL Shortener with Real-Time Analytics

Django · DRF · Redis · Celery · PostgreSQL · Docker Compose · JWT

Architected a containerized URL shortening platform with JWT-secured REST APIs, Redis caching, and async Celery workers. Delivered real-time click analytics with device, browser, and geographic breakdowns plus CSV/PDF export, achieving <200ms API response times and 10k+ concurrent user capacity via Docker Compose orchestration.

AI Resume Analyzer — Full-Stack SaaS Platform

Django · DRF · React 19 · PostgreSQL · Google Gemini · scikit-learn · PyMuPDF · Docker

Built a full-stack ATS optimization platform combining TF-IDF cosine similarity scoring with Google Gemini- powered improvement suggestions. Engineered PDF ingestion via PyMuPDF, matched/missing keyword detection, ReportLab report export, and a token-refresh retry queue in React Axios interceptors — deployed as decoupled Docker containers with PostgreSQL persistence.

Bank Customer Churn Prediction

Python · pandas · scikit-learn · XGBoost · SMOTE · Matplotlib

Built an end-to-end ML pipeline to predict bank customer churn. Applied SMOTE for class imbalance, engineered features with ColumnTransformer, and tuned an XGBoost classifier achieving 86.71% test accuracy and ROC- AUC of 0.85 with minimal overfitting (1.41% train-test gap).

Blood Donation Management System

Django · Python · HTML · Oracle — Full-Stack Web Application

Developed a full-stack Django web application for managing blood donors, requests, and inventory. Implemented donor registration, blood group filtering, request tracking, and admin management using Django ORM, template rendering, and role-based access control.

Heart Failure Prediction

Python · scikit-learn · pandas — Classification · Feature Engineering · Model Evaluation

Developed binary classification models on structured healthcare data using feature engineering, hyperparameter tuning, and performance benchmarking with precision, recall, F1-score, and ROC-AUC to identify reliable diagnostic signals.

Real-Time Financial Anomaly Detection

Python · scikit-learn · Isolation Forest · PCA · Streamlit · pandas

Built an unsupervised ML pipeline to detect fraudulent credit card transactions (492 frauds / 284,807 transactions) using Isolation Forest without labeled training. Applied RobustScaler and PCA for 2D anomaly visualization. Deployed as a real- time Streamlit dashboard for live inference.

EXPERIENCE

Backend Developer Intern

Enigmatix, Bahawalpur

6 months

On-site

Reduced manual processing time by 30% by engineering 5+ Django REST API modules and CRUD systems for business workflows using Python, SQLite, and Oracle.

Improved database query performance by 25% by designing and optimizing Django ORM models, enhancing backend data reliability and response speed.

Cut post-deployment defects by 20% by leading systematic API debugging and enforcing Git/GitHub version control best practices.

Designed and maintained database models to support scalable backend features and cleaner application logic.

TECHNICAL SKILLS

Programming Python, SQL

Machine Learning scikit-learn, XGBoost, regression, classification, SMOTE, model evaluation, feature engineering

Data Analysis pandas, NumPy, Matplotlib, EDA, data cleaning, feature engineering

Backend Django, Django REST Framework, REST APIs, CRUD development, SQLite, Oracle, PostgreSQL

Frontend React 19, Vite, TailwindCSS, Axios, React Router

DevOps / Tools Docker, Docker Compose, Redis, Celery, Git, GitHub, Gunicorn

CERTIFICATIONS

- Machine Learning with Python (IBM)
- Exploratory Data Analysis for Machine Learning (IBM)
- Data Analysis with Python (IBM)
- Databases and SQL for Data Science (IBM)
- Statistics for Data Science with Python (IBM)
- AI For Everyone (DeepLearning.AI)

LANGUAGES

Urdu • English